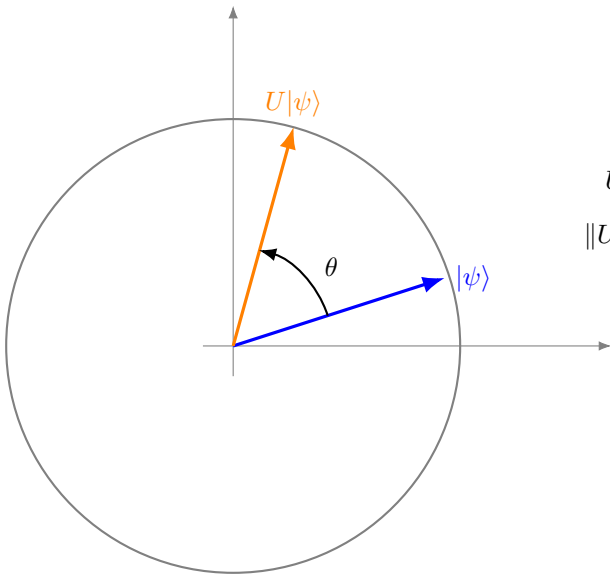




$$U^*U = I$$

$$\|U\psi\| = \|\psi\|$$